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R2R: Reasoning-to-Residual Alignment for Financial Time Series Forecasting via Fine-Tuning LLMs

(/pdf?id=MljtMieSB4)

KDD 2026 Research Track Cycle 2 Submission2831 Authors

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Keywords: Time Series Forecasting, Large Language Models, Fine-tuning, Financial reasoning, Residual Alignment

Abstract:
Financial time series forecasting remains a formidable challenge due to the disconnect between endogenous historical trends and exogenous macroeconomic shocks. While state-of-the-art channel-independent transformers excel at capturing long-term temporal dependencies, they remain semantically blind to regime-shifting events like monetary policy adjustments. Existing multimodal approaches often fail to bridge this gap, as naive fusion mechanisms frequently lead to modality interference, where high-dimensional textual noise obscures the underlying technical signals. To address this challenge, we propose a novel Reasoning-to-Residual Alignment framework that reformulates the integration of economic logic as a residual learning task. Our framework decouples the forecasting process into two components: a technical backbone that generates a baseline numerical trajectory, and the LLM that functions as a specialized residual predictor. Crucially, the LLM is fine-tuned not on raw price generation, but on regressing the forecast residual (i.e., the residual between the baseline and ground truth) conditioned on macro-event textual data. Such alignment enables the LLM to explicitly learn the quantitative mapping from semantic news to the volatility corrections required by the technical model. The final forecast is derived by rectifying the baseline momentum with the LLM-predicted residual. Extensive experiments on financial datasets demonstrate that our approach significantly reduces Mean Squared Error (MSE) compared to vanilla baselines, proving that fine-tuned semantic models can effectively calibrate the blind spots of numerical forecasters.

Resubmission Flag: No
Artifact Pledge: Yes
Submission Guidelines Blind: Yes
Submission Guidelines Format: Yes
Submission Guidelines Limit: Yes
Submission Guidelines Authorship: Yes
Submission Number: 2831

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Official Review of Submission2831 by Reviewer sMqD

Edit

Official Review by Reviewer sMqD (Huaming Chen (/profile?id=~Huaming_Chen1)) 24 Mar 2026, 18:43

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer sMqD

Paper Summary:

In this paper, a framework called R2R is presented to decomposes financial time series forecasting into two components to enhance the performance on financial time series task. The framework is evaluated on seven assets with hourly data, showing improvements over baselines in forecasting accuracy and trading performance.

Paper Strengths:

1. The separation of endogenous momentum from exogenous shock correction is intuitive and mirrors how practitioners actually think about markets. The multiplicative fusion formulation is a sensible design choice, during quiet periods the residual approaches zero, preserving the backbone's predictions, while during events the LLM injects corrections. This is a cleaner design than naive concatenation-based fusion.
2. The causal safety buffer and the ex-ante bias calculation demonstrate awareness of a common pitfall in financial ML research. The strictly chronological train/val/test split is appropriate.
3. The paper evaluates across seven assets spanning two distinct market types (crypto and equities), multiple forecast horizons, and addresses multiple research questions including ablation, policy-event-specific analysis, trading simulation, and computational efficiency. The policy event impact analysis (Table 3) is particularly informative.
4. The trading simulation (Table 4) adds practical grounding. R2R being the only model with a non-negative Sharpe ratio on BTC is a notable result. The efficiency analysis showing competitive latency despite the LLM backbone is also useful.

Paper Weaknesses:

1. The paper contains inconsistencies (conference template says "Conference'17, July 2017" throughout), and some claims are overstated (e.g., "fundamentally reframes" appears twice). The notation is sometimes inconsistent (R* vs R*_t, inconsistent use of subscripts). The paper also uses "PQ6" instead of "RQ6" in Section 6.
2. The paper justifies multiplicative over additive fusion with one sentence ("news mainly strengthens or weakens that trajectory rather than shifting it by a fixed amount"). No empirical comparison between additive and multiplicative fusion is provided. For assets with prices near zero or during extreme events, multiplicative corrections could produce unstable outputs.
3. The paper states the framework is "event-triggered" in the limitations, but the main results in Table 1 appear to report overall performance. It is unclear whether Table 1 evaluates only on event-window samples or on all test samples. If only event windows, the practical utility is limited to ~311 test events. If all samples, then R2R should behave identically to the backbone during non-event periods, and the reported gains would be diluted — raising questions about how meaningful the improvements truly are across different evaluation scopes.
4. No error bars, confidence intervals, or significance tests are reported anywhere. Financial time series are notoriously noisy, and the improvements (e.g., MSE differences in the fourth decimal place) could easily fall within random variation. Without multiple runs or bootstrap confidence intervals, it is impossible to assess whether the reported gains are reliable.
5. The paper does not compare against CAMEF [27], which is explicitly discussed in related work as a recent multimodal method integrating news with price data. Other missing comparisons include TimesNet [21] (mentioned in related work), and simpler sentiment-score-based approaches that would help isolate whether the LLM reasoning is truly necessary versus a basic sentiment signal. The LLM-based baseline group contains only Time-LLM, which is insufficient.

Resubmission:

NA

Repeat Reviewer: No

Questions And Suggestions For Rebuttal:

1. Table 2 anomaly: Why do "Gemini-v3 Only" and "GPT-V5 Only" appear under "Standard Technical Baseline," and why does Gemini-v3 Only (MSE 0.000711) outperform the full R2R framework (MSE 0.000768)? Please clarify what these baselines are and how they were evaluated.

2. Evaluation scope of Table 1: Are the results in Table 1 computed over all test set timestamps, or only over event-window timestamps? If the former, what fraction of test timestamps fall within event windows, and what is the performance breakdown between event and non-event periods?
3. Teacher LLM quality: Can the authors provide a quantitative comparison of the denoised residuals R* versus the raw residuals R_raw as training targets (e.g., correlation with ground truth, downstream forecasting accuracy when using each)? How sensitive is the framework to the choice of teacher LLM?
4. Statistical reliability: Can the authors report results over multiple random seeds or bootstrap confidence intervals for the main results? Specifically, are the improvements in Table 1 statistically significant at $p < 0.05$?
5. Additive vs. multiplicative fusion: Can the authors provide an ablation comparing additive fusion against the proposed multiplicative formulation? What happens when the baseline prediction is near zero?
6. Missing baselines: Can the authors include comparisons with CAMEF [27] and a simple sentiment-score baseline (e.g., using a pre-trained financial sentiment model to produce a scalar correction) to isolate the contribution of structured chain-of-thought reasoning versus basic directional sentiment?
7. Generalization to unseen event types: How does R2R perform on macro events not represented in the training set? Have you tested on event categories held out during training?
8. Ablation across assets: The ablation (Table 2) is limited to ETH. Do the same trends hold for equities (SPX, INDU) where the improvements in Table 1 are smaller? The relative contribution of each component may differ across asset classes.
9. Scalability of event curation: The current framework requires manually defined event categories and curated event streams. How would R2R scale to real-time deployment where event detection and classification must also be automated?
10. Negative results in Table 3: Several entries show negative MAE reduction (e.g., NDX under Retail at -15.33%, DOGE under C.CPI at -17.39%). These cases where R2R hurts performance deserve explanation. Under what conditions does the semantic correction degrade accuracy, and can the framework detect and suppress corrections in these cases?

Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community
Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.
Technical Quality: 2: Low - The paper has several technical weaknesses, such as minor methodological flaws, insufficient analysis, or unsupported conclusions. While the work shows some level of competence, it lacks thoroughness and precision. Improvements are necessary for it to be considered robust.
Presentation: 2: Low - The paper has noticeable issues with clarity and coherence. The writing may contain several grammatical and typographical errors. Figures and tables are present but may not be well-integrated or effectively used. The presentation allows for understanding but requires effort from the reader.
Reproducibility: 2: Low - The paper includes some information about the methods, data, and procedures, but key details are missing. There may be supplementary materials, but they are incomplete or unclear. Reproducing the results would require significant effort and additional information.
Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.
Ethics Review Flag: No
Ethics Review Description: NA
Llm Usage Description: I have use LLMs to refine the wording.

Official Review of Submission2831 by Reviewer iDqU

Official Review by Reviewer iDqU 19 Mar 2026, 13:01

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer iDqU

Paper Summary:
This paper proposes a Reasoning-to-Residual (R2R) framework that integrates time series forecasting models with large language models (LLMs) for financial forecasting. The framework separates the prediction process into a technical backbone that produces baseline forecasts from historical time series data and an LLM that predicts residual corrections based on macroeconomic events. By decomposing forecasting into endogenous trends and

exogenous residuals, the approach aims to incorporate semantic information that traditional time series models cannot capture. The paper also introduces an LLM-guided residual construction and denoising mechanism that transforms raw residual signals into smoothed and logically grounded vectors.

Paper Strengths:

1. This work presents a novel forecasting decomposition strategy that distinguishes between endogenous and exogenous components. It establishes a clear framework that separates price dynamics from macro-event-driven shocks. Such a decomposition offers an effective approach to integrate textual information into time-series forecasting.
2. Instead of using LLMs for direct numerical forecasting, the method predicts residual corrections, which reduces the risk of arithmetic errors. This formulation aligns better with the strengths of LLMs in semantic reasoning.
3. The paper presents the proposed method with a clear and well-organized pipeline, and the technical details are described thoroughly.

Paper Weaknesses:

1. The paper introduces a teacher LLM-based residual denoising step, but the rationale and theoretical justification remain unclear. It is not fully explained why an LLM would reliably remove stochastic market noise from residual signals.
2. The number of baseline models is relatively limited, and the ablation study is conducted on a single dataset; it would be more convincing if similar analyses were performed on other datasets.

Resubmission:
NA

Repeat Reviewer: No
Questions And Suggestions For Rebuttal:

1. Could the authors provide additional analysis or experiments to demonstrate the effectiveness of the residual predictor under previously unseen macro events or different market conditions?
2. The ablation study is conducted on a single dataset. While the results provide useful insights, it is unclear whether the observed improvements remain consistent across other datasets or market environments. Could the authors provide additional ablation results on other datasets to demonstrate the stability and generality of each component in the proposed framework?

Relevance: 4: High - The work is relevant to the Research track of KDD and is of broad interest to the community
Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.
Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.
Presentation: 3: Moderate - The paper is organized and generally clear. The writing is mostly free of grammatical and typographical errors, making it easy to read. Figures and tables are effectively used to support the text. The presentation facilitates understanding and conveys the key points effectively.
Reproducibility: 2: Low - The paper includes some information about the methods, data, and procedures, but key details are missing. There may be supplementary materials, but they are incomplete or unclear. Reproducing the results would require significant effort and additional information.
Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.
Ethics Review Flag: No
Ethics Review Description: NA
Llm Usage Description: NA

Official Review of Submission2831 by Reviewer XWnP

Official Review by Reviewer XWnP 19 Mar 2026, 07:39

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer XWnP

Paper Summary:

This paper proposes R2R, a "Reasoning-to-Residual" alignment framework for financial time series forecasting that decouples prediction into an endogenous numerical prior from a standard time-series backbone and an exogenous residual predicted by a fine-tuned LLM from macro-event text.

Paper Strengths:
The paper addresses an important, practically relevant challenge: integrating macroeconomic narratives into time-series forecasting via a conceptually clean and well-motivated residualization framework. The idea of aligning LLM reasoning to residual corrections is appealing.

Paper Weaknesses:
The current experimental evidence has notable gaps:

- absence of direct comparisons to the closest multimodal/event-driven baselines (e.g., TEST, CAMEF)
- limited analysis of multiplicative vs. additive/log-space formulations
- lack of significance testing

Resubmission:
NA

Repeat Reviewer: No
Questions And Suggestions For Rebuttal:

- how does a simple residual regressor (e.g., RoBERTa embeddings + MLP, or gradient-boosted trees on structured event features) perform?
- can you compare teacher LLM denoising against simple statistical smoothers and show generalization differences?

Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community
Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.
Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.
Presentation: 2: Low - The paper has noticeable issues with clarity and coherence. The writing may contain several grammatical and typographical errors. Figures and tables are present but may not be well-integrated or effectively used. The presentation allows for understanding but requires effort from the reader.
Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.
Reviewer Confidence: 2: Low - The reviewer has some knowledge of the subject area and is somewhat familiar with the research methods and context of the paper. They understand the main points but may lack depth in certain areas. The reviewer is reasonably confident in their assessment but acknowledges some limitations in their expertise.
Ethics Review Flag: No
Ethics Review Description: NA
Llm Usage Description: LLM is used to help identify the research gap in this paper

Official Review of Submission2831 by Reviewer CFnK

Official Review by Reviewer CFnK 09 Mar 2026, 02:59 (modified: 09 Mar 2026, 03:07)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer CFnK
Revisions (/revisions?id=ocexQPdmjc)

Paper Summary:
The paper proposes a novel framework named R2R, Reasoning-to-Residual Alignment for event-driven financial forecasting. The research aims to propose an output-level fusion mechanism that can quantify the magnitude of the textual events to the financial time-series trends. Specifically, the proposed framework is composed of a pure time-series model which functions as a technical ``baseline'', and a teacher LLM (Gemini) is prompted to denoise the raw residual between the ground-truth future trajectory and the baseline forecast, producing a structured

target containing both the semantic reasoning and the rectified residual vector; and then, a smaller LLM is fine-tuned to predict residual target, and the predicted residual is combined with the forecasted time-seres from the pure time-series baseline. The experiments show that the proposed method achieves strong improvements over previous single-modality and multi-modality forecasting methods on most cases, and the ablation studies suggest that the proposed pipeline is generally reasonable.

Paper Strengths:

1. The paper studies a well-defined research problem, where in the financial market, some salient macro events may significantly affect the market trends, while pure time-series models are often semantically blind to such exogenous shocks
2. The proposed method is generally reasonable. The model first forecasts a time-series as the endogenous baseline, and then estimates the potential event shock by leveraging LLMs through a residual prediction manner
3. The experiment results show that R2R achieves strong performances against several previous forecasting baselines, and the paper also provides ablation tests and case studies to support the proposed design

Paper Weaknesses:

1. The presentation of the paper is kind of a problem. Section 4 seems somewhat redundant, where there are just a few short paragraphs to overview the model, while most of the actual terminologies and technical definitions are introduced later in Section 5
2. The technical novelty seems somewhat limited. Although the paper proposes the time-series baseline model, teacher-LLM denoising, and multiplicative calibration, the overall framework still looks more like a multiplicative merge of the two forecasting results from time-series forecster and the LLM forecaster.
3. Regarding R2R, some parts are still not well explained. In Stage 2, Gemini is prompted with R_{raw} to produce both the reasonings and the "vector rectification," $R*$. Then, in Stage 3, a smaller LLM is trained to forecast $R*$ using the mean residual in Equation 6 as input. Therefore, the ultimate objective seems to be forecasting $R*$. If so, what is the actual purpose of collecting the reasonings? How exactly do these reasonings help the forecasting performance? I also think the paper should verify this through an ablation study, for example by comparing training with reasonings and training without reasonings.
4. It would be better to showcast some actual reasoning outputs by the teacher or student LLMs.
5. The claims in the two paragraphs "Arithmetic Hallucination in Direct Generation" and "Modality Misalignment in Embedding Fusion" lack references.
6. Critical parameters seem missing, like the training epochs, lora settings, learning rates, etc.

Resubmission:
Not a resubmission

Repeat Reviewer: No
Questions And Suggestions For Rebuttal:

1. Could the authors provide abaltion tests with or without the reasonings from Gemini? How about just train the LLM to forecast $R*$ from residual b_t and event E_t
2. Could the authors showcase some reasonings from Gemini?
3. Why use Gemini to generate a $R*$ as the training target? why not just use R_{raw}? How is R_{raw} different from $R*$ can authors showcast the generated $R*$s or make some statistics on comparion between R_{raw} and $R*$
4. What are the training paramters? e.g. training epochs, lora settings they seem all missing.

Relevance: 4: High - The work is relevant to the Research track of KDD and is of broad interest to the community
Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.
Technical Quality: 2: Low - The paper has several technical weaknesses, such as minor methodological flaws, insufficient analysis, or unsupported conclusions. While the work shows some level of competence, it lacks thoroughness and precision. Improvements are necessary for it to be considered robust.
Presentation: 2: Low - The paper has noticeable issues with clarity and coherence. The writing may contain several grammatical and typographical errors. Figures and tables are present but may not be well-integrated or effectively used. The presentation allows for understanding but requires effort from the reader.
Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently

documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.

Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.

Ethics Review Flag: No

Ethics Review Description: No

Llm Usage Description: 🤖 revise my gramma

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